SHINDENGEN

Schottky Rectifiers (SBD)

Single

D5S9M

90V 5A

FEATURES

- Tj150℃
- P_{RRSM} avalanche guaranteed
- Fully Isolated Molding

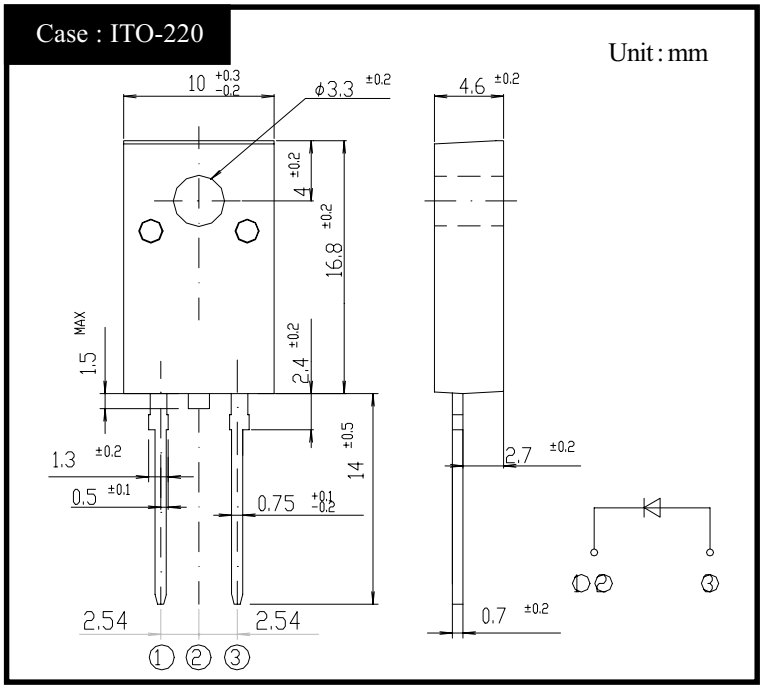
APPLICATION

- Switching power supply
- DC/DC converter
- Home Appliances, Office Equipment
- Telecommunication

OUTLINE DIMENSIONS

Case : ITO-220

Unit : mm



RATINGS

●Absolute Maximum Ratings (If not specified Tc =25℃)

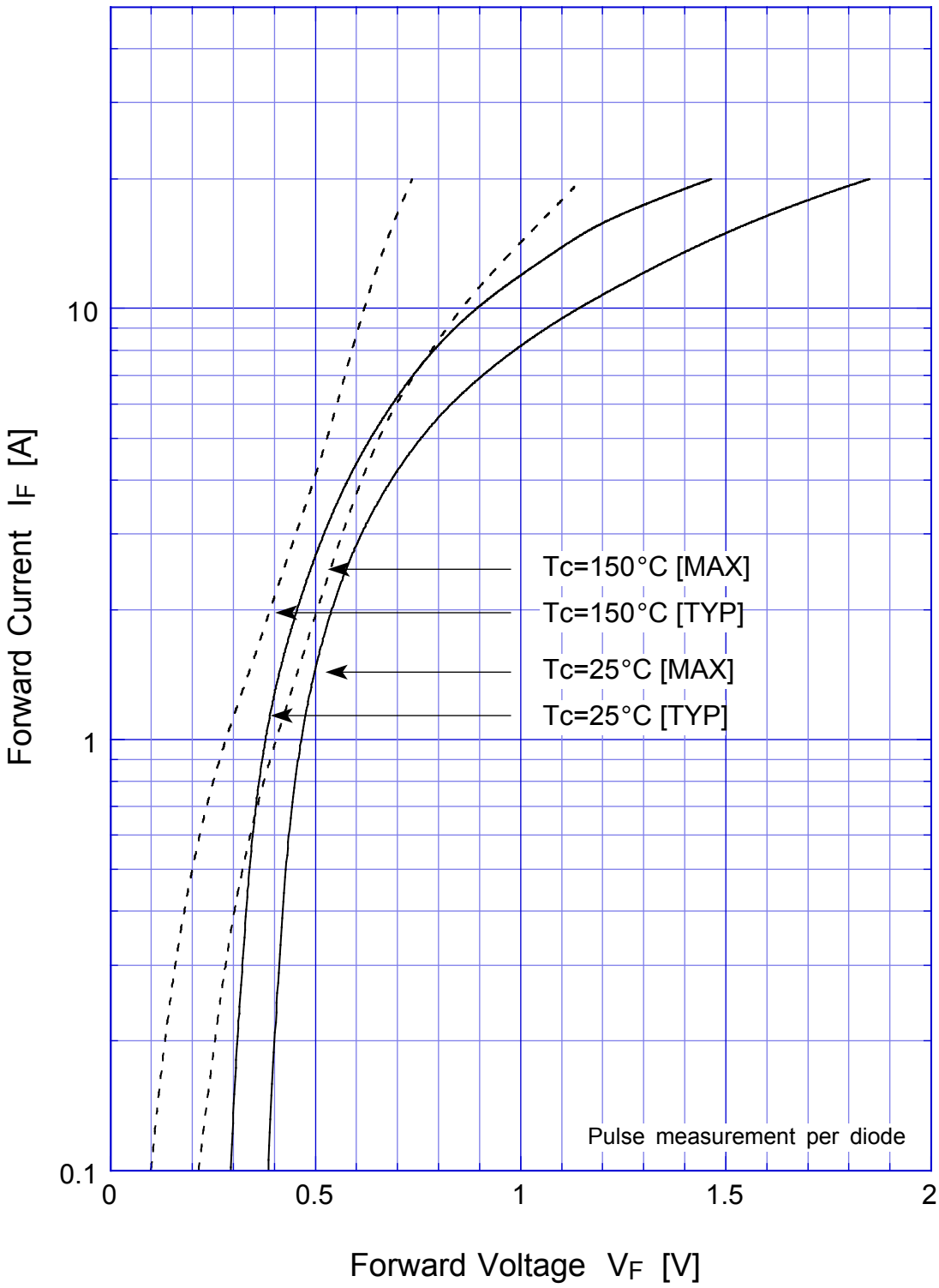
Item	Symbol	Conditions	Ratings	Unit
Storage Temperature	Tstg		−40～150	℃
Operating Junction Temperature	Tj		150	℃
Maximum Reverse Voltage	V_{RM}		90	V
Repetitive Peak Surge Reverse Voltage	V_{RRSM}	Pulse width 0.5ms, duty 1/40	100	V
Average Rectified Forward Current	I_O	50Hz sine wave, R-load, Tc=121℃	5	A
Peak Surge Forward Current	I_{FSM}	50Hz sine wave, Non-repetitive 1 cycle peak value, Tj=125℃	100	A
Repetitive Peak Surge Reverse Power	P_{RRSM}	Pulse width 10 μ s, Tj=25℃	330	W
Dielectric Strength	Vdis	Terminals to case, AC 1 minute	1.5	kV
Mounting Torque	TOR	(Recommended torque: 0.3N･m)	0.5	N･m

●Electrical Characteristics (If not specified Tc =25℃)

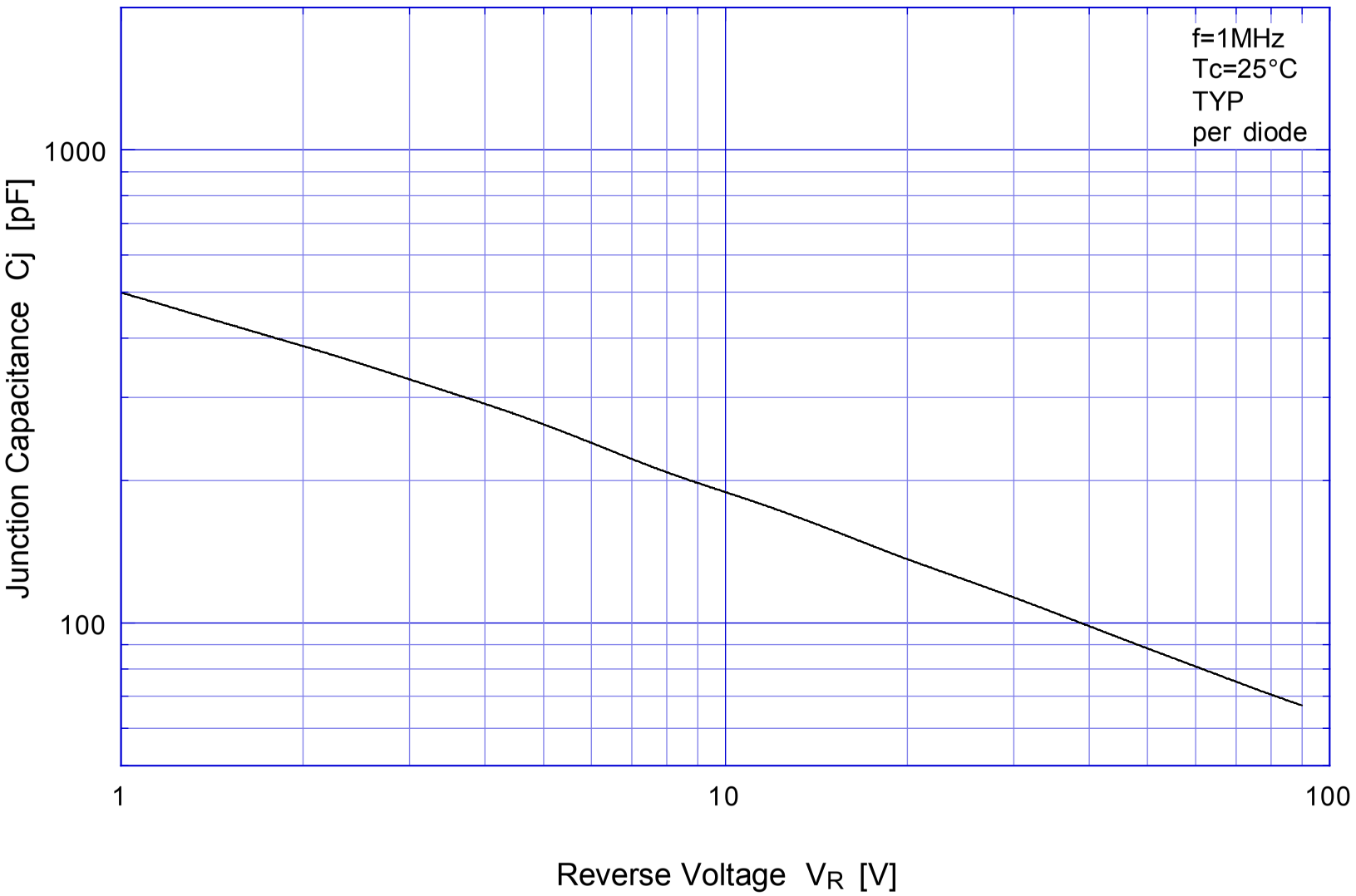
Item	Symbol	Conditions	Ratings	Unit
Forward Voltage	V_F	I_F=5A, Pulse measurement	Max 0.75	V
Reverse Current	I_R	V_R=V_{RM}, Pulse measurement	Max 3.0	mA
Junction Capacitance	Cj	f=1MHz, V_R=10V	Typ 185	pF
Thermal Resistance	θ jc	junction to case	Max 4.5	℃/W
	θ jf	junction to heatsink	Max 6.0	

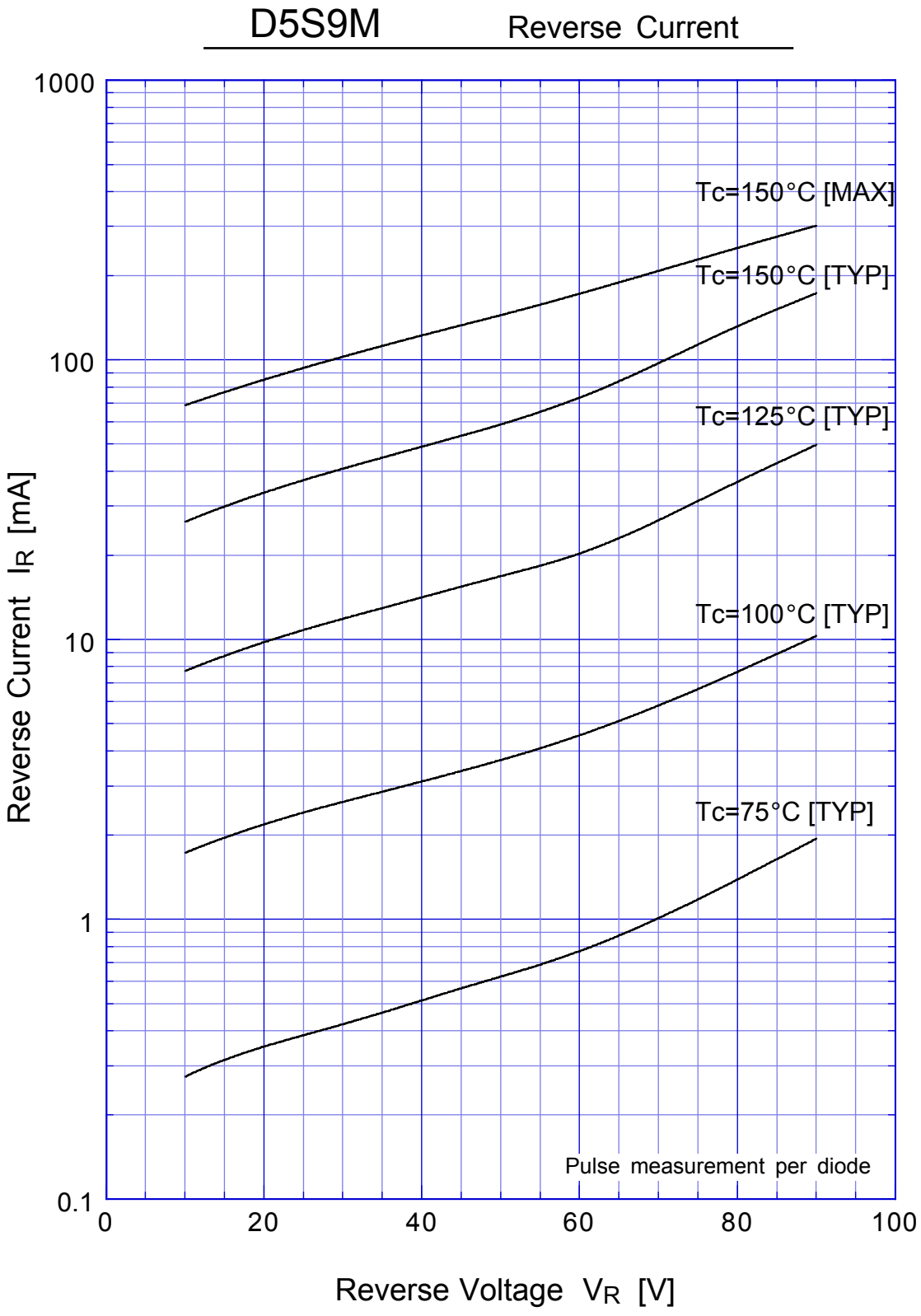
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D5S9M Forward Voltage

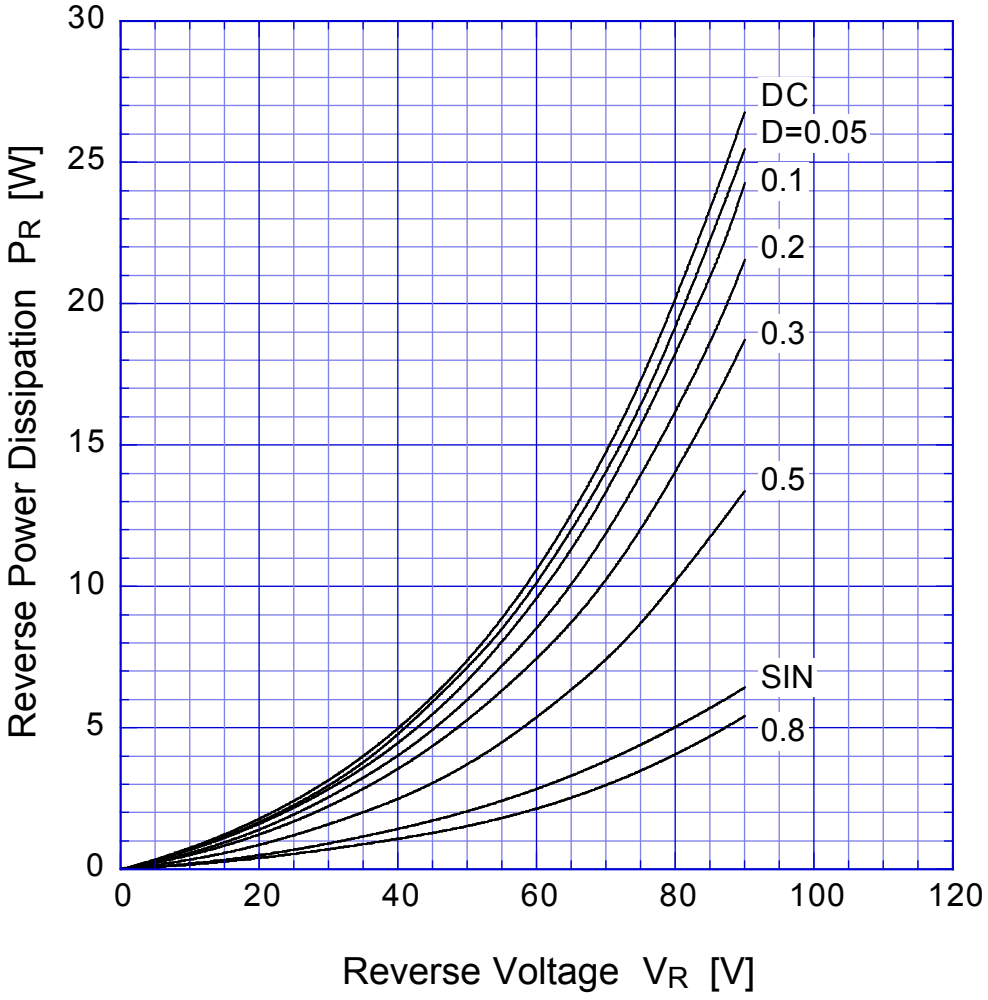


D5S9M Junction Capacitance

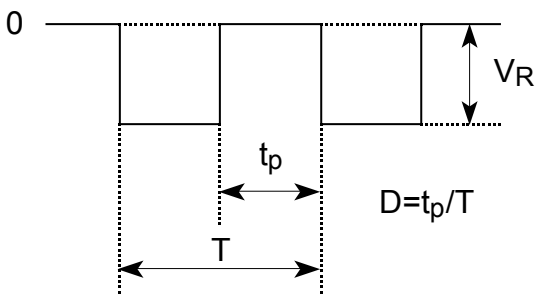



D5S9M
Reverse Current
1000
100
10
1
0.1
Reverse Current IR [mA]
Tc=150°C [MAX]
Tc=150°C [TYP]
Tc=125°C [TYP]
Tc=100°C [TYP]
Tc=75°C [TYP]
Pulse measurement per diode
0
20
40
60
80
100
Reverse Voltage VR [V]

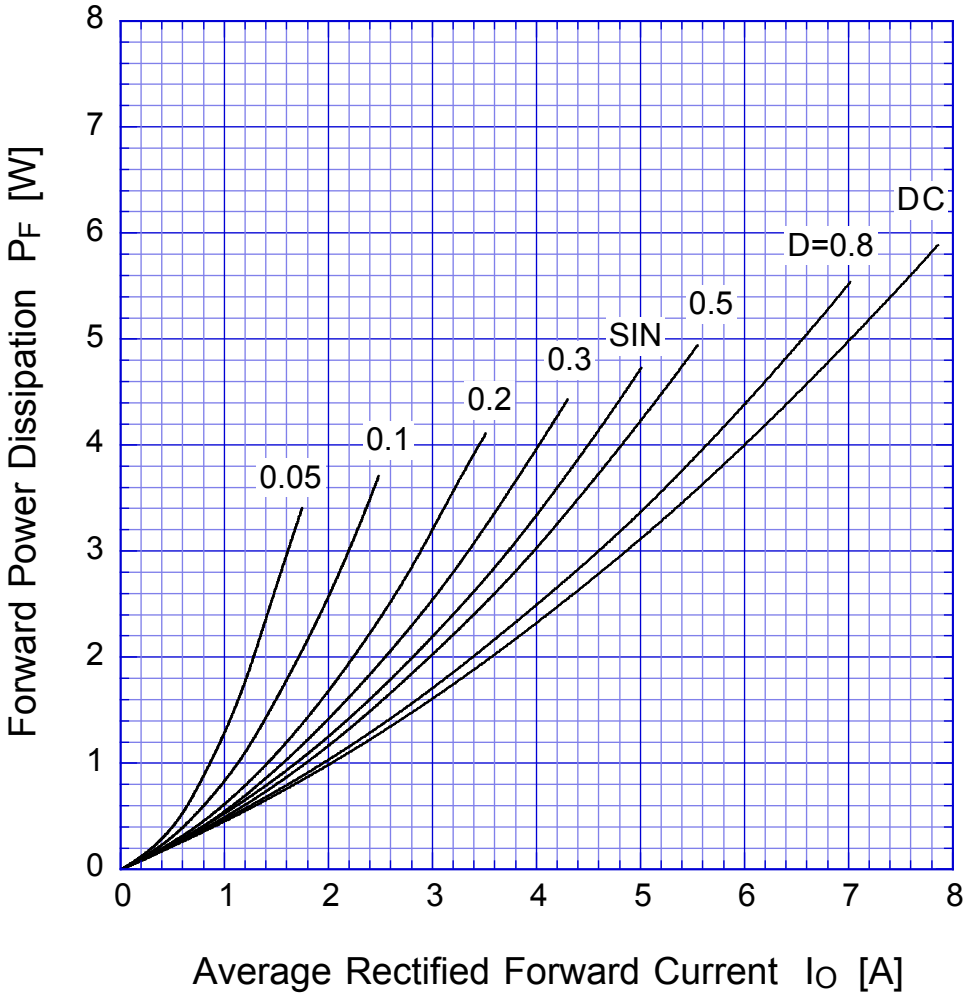
D5S9M Reverse Power Dissipation



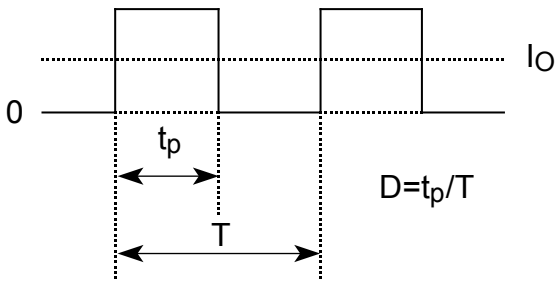
Tj = 150°C



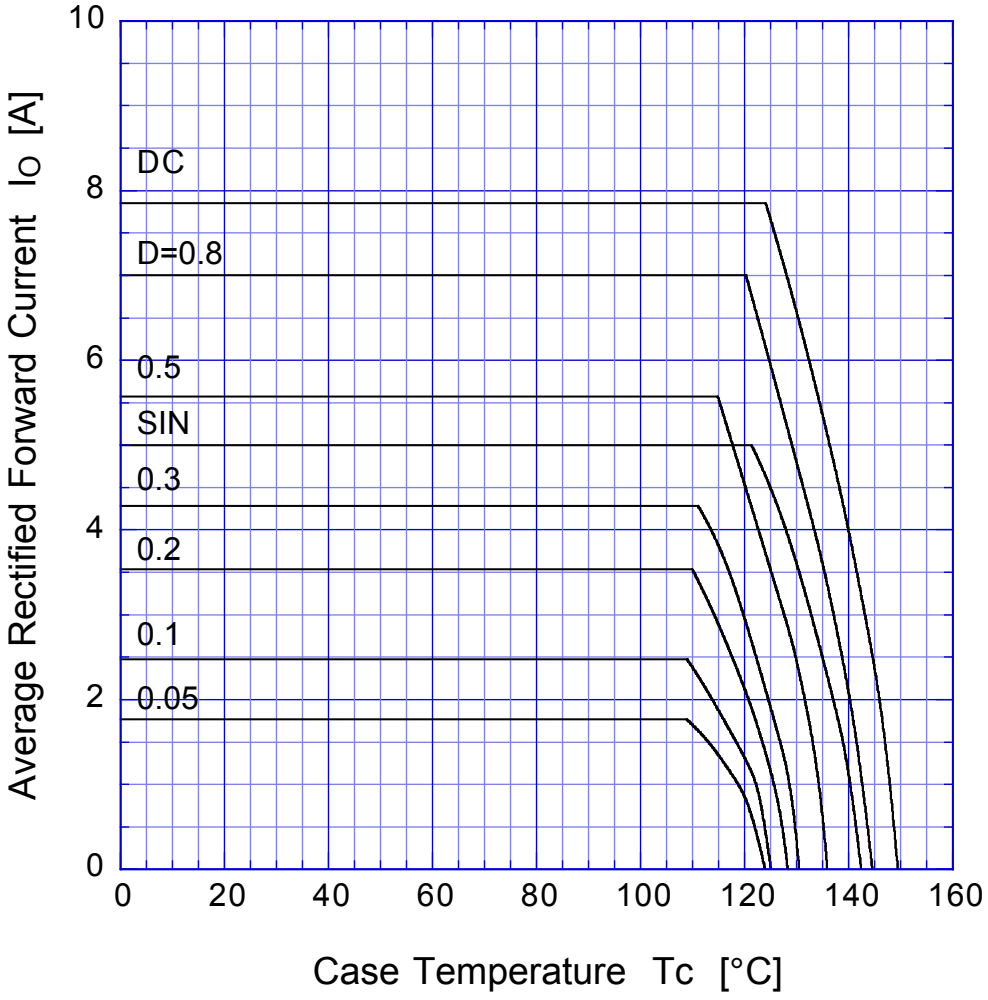
D5S9M Forward Power Dissipation



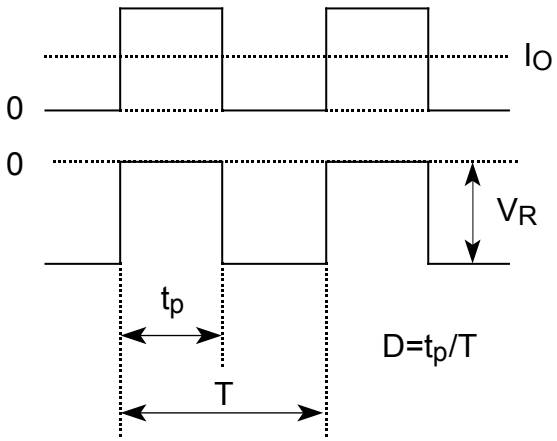
Tj = 150°C



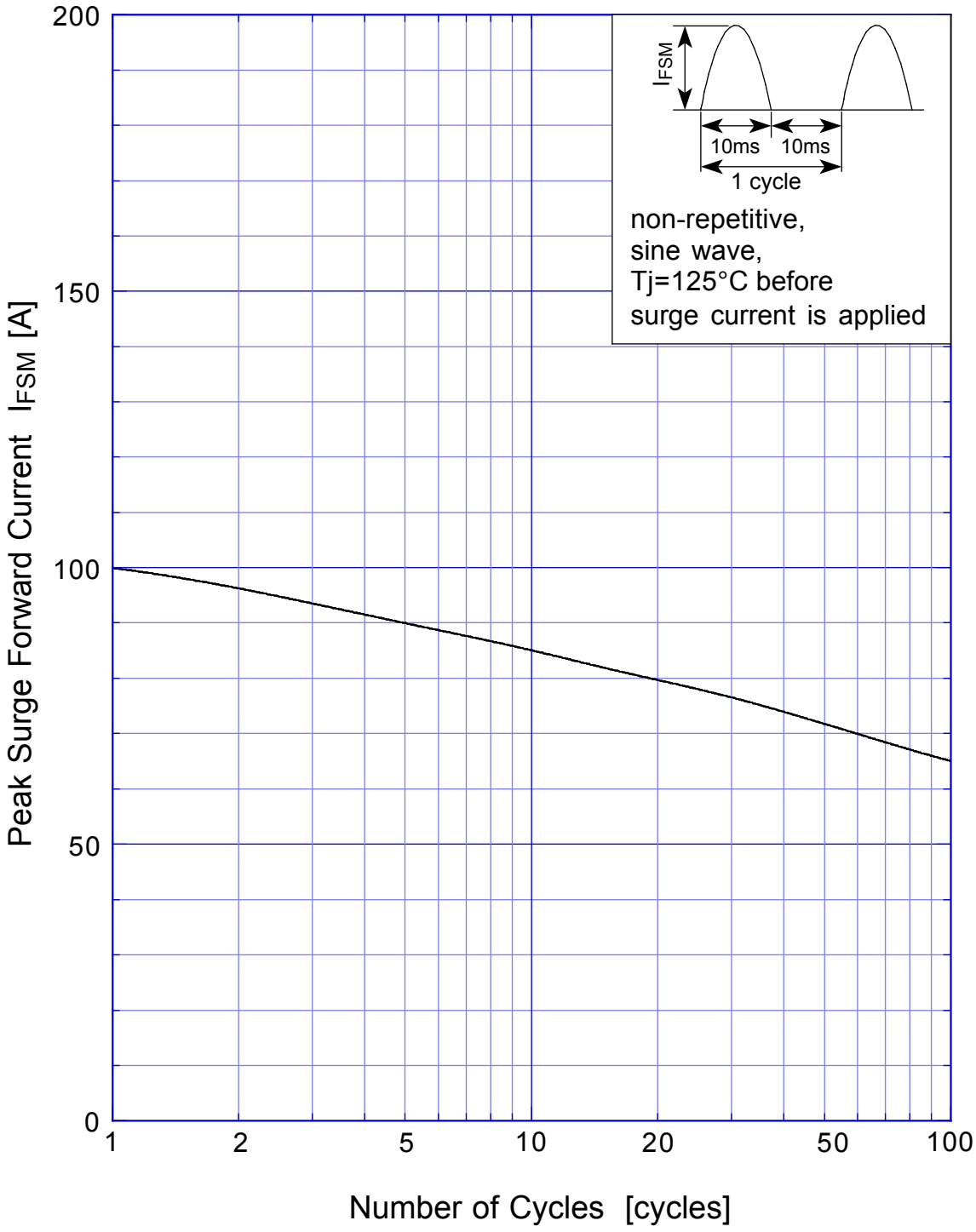
D5S9M Derating Curve



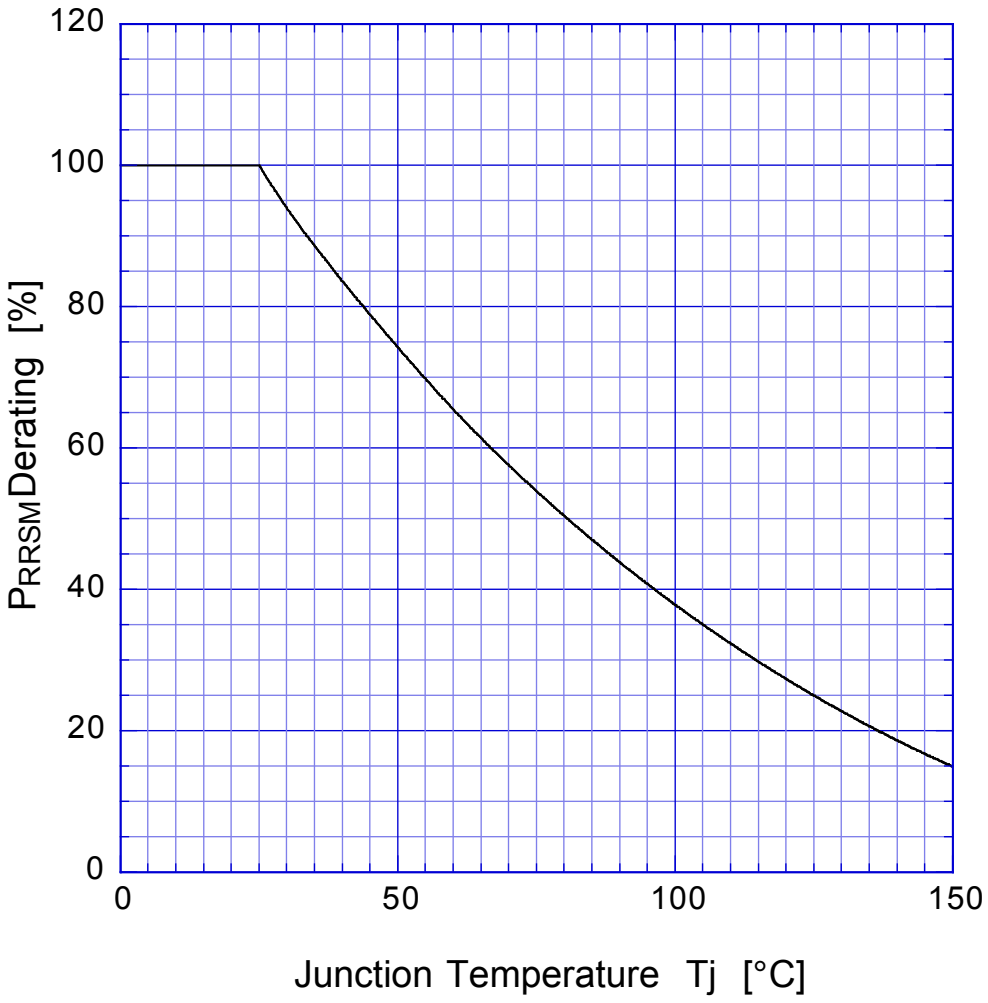
V_R = 45V

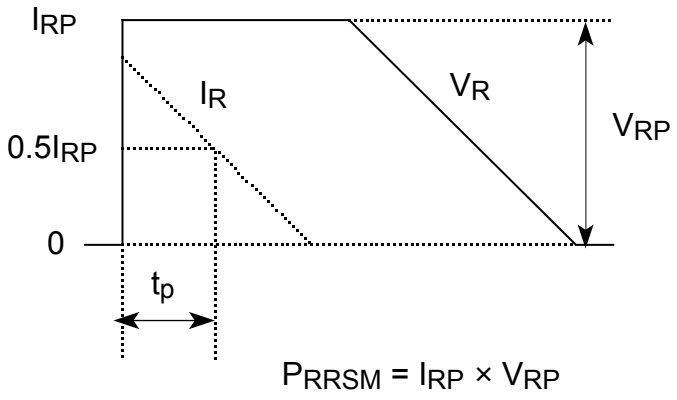


D5S9M Peak Surge Forward Capability

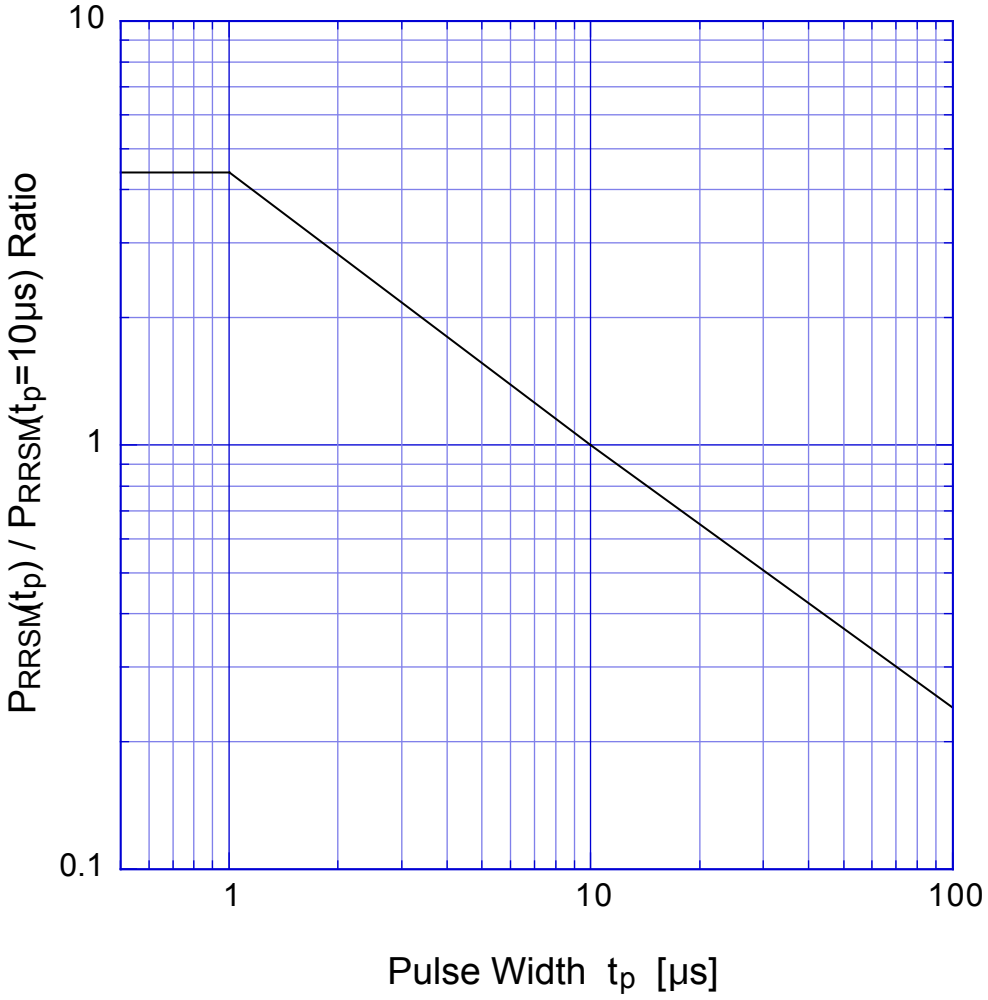


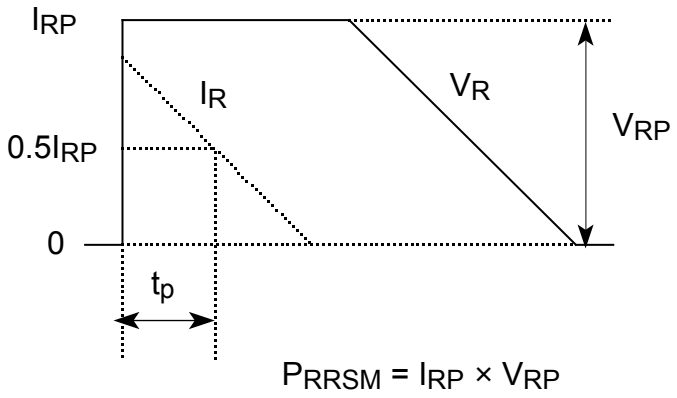
SBD Repetitive Surge Reverse Power Derating Curve





SBD Repetitive Surge Reverse Power Capability





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Datasheets for electronics components.